

Applied Statistics

Course Project

Heart Failure Analysis



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Abstract

Cardiovascular diseases (CVDs) remain the leading cause of death worldwide, accounting for 17.9 million fatalities annually. The early and accurate detection of heart disease is crucial for reducing mortality and improving patient outcomes. Traditional methods like angiography, while precise, are invasive, costly, and limited in availability, especially in developing regions. Recent advancements in machine learning offer promising alternatives, leveraging non-invasive approaches for prediction and diagnosis. This project analyzes the Heart Failure Prediction dataset, which combines data from multiple sources to identify significant predictors of heart disease, such as age, cholesterol, and exercise-induced angina. By utilizing logistic regression and diagnostic tools like Variance Inflation Factor (VIF) and Cook's Distance, the study develops a predictive model. Insights from this analysis contribute to the growing body of evidence supporting the integration of computational techniques in medical decision-making.

Introduction

Cardiovascular diseases (CVDs) are the leading cause of death worldwide, responsible for approximately 17.9 million deaths annually, representing 31% of all global fatalities[2]. The significant health and economic burden posed by CVD has intensified the need for effective prevention, early detection, and management strategies[1]. Factors such as high blood pressure, cholesterol, and physical inactivity are major contributors, with high blood pressure being the most prominent risk factor, as evidenced by extensive research[4]. Traditional diagnostic methods like angiography, while precise, are invasive, costly, and inaccessible in many regions, particularly in resource-limited settings[1]. These challenges motivate the exploration of alternative, non-invasive diagnostic approaches that can be implemented on a larger scale[5,6].

Machine learning (ML) has emerged as a promising solution, offering non-invasive, efficient, and scalable methods for analyzing complex medical data[5]. ML models, including Random Forest (RF), Support Vector Machine (SVM), and boosting algorithms, have demonstrated their ability to accurately predict CVD. For instance,[5] found RF to be the most effective algorithm for classifying coronary artery disease, achieving an accuracy of 92.9%, followed by SVM at 89.7%. Similarly, a meta-analysis [6]evaluating over 3 million patients demonstrated the high predictive ability of ML algorithms, with SVM achieving a pooled AUC of 0.92 for cardiovascular outcomes. These findings underscore the potential of ML to enhance diagnostic accuracy while addressing the limitations of traditional methods [5, 6].

The Heart Failure Prediction dataset, which combines information from five reputable sources, Cleveland, Hungarian, Switzerland, Long Beach, VA, and Statlog, is used in this study. The dataset consists of 918 observations and 11 characteristics that are commonly known to be important predictors of heart disease [3], including age, cholesterol, kind of chest pain, and exercise-induced angina. As demonstrated in a study[5], the inclusion of datasets increases the generalizability of prediction models in addition to improving data variety. Furthermore, the dataset's public availability through the UCI Machine Learning Repository ensures transparency and supports reproducibility. This project aims to address the problem of early detection of heart disease by applying

logistic regression to identify significant predictors and evaluate their impact on disease risk.

Problem Statement

The objective is to identify significant predictors of heart disease and develop an interpretable logistic regression model to facilitate early detection of cardiovascular disease (CVD).

Methodology

This study analyzed the Heart Failure Prediction dataset, which consolidates data from five reputable sources: Cleveland Clinic Foundation, Hungarian Institute of Cardiology, Switzerland University Hospital, Long Beach VA Medical Center, and the Statlog project. The dataset includes 918 observations and 11 attributes, such as age, cholesterol levels, chest pain type, and exercise-induced angina, which are widely recognized as critical predictors of heart disease.

Data Preprocessing

Data preprocessing was conducted to prepare the dataset for logistic regression analysis. Categorical variables, such as **Sex** and **ChestPainType**, were encoded into dummy variables to enable their use in the regression model. For instance, **ChestPainType** was converted into binary indicators representing its four categories: typical angina, atypical angina, non-anginal pain, and asymptomatic. Continuous predictors, such as age, cholesterol, and resting blood pressure, were normalized to improve model convergence and interpretability by ensuring variables operated on similar scales.

Model Fitting

Logistic regression was chosen for its interpretability and ability to estimate the relationship between predictors and the binary outcome variable **HeartDisease** (1 = presence, 0 = absence). The model was specified as:

$$\log it(\pi) = \beta_0 + \beta_1 (Age) + \beta_2 (RestingBP) + \beta_3 (Cholesterol) + \beta_4 (Sex) + \beta_5 (ChestPainType)$$

where π represents the probability of heart disease.

Diagnostic Checks

Diagnostic evaluations were performed to validate the model. Variance Inflation Factor (VIF) was calculated to assess multicollinearity among predictors, ensuring that none were overly correlated. Residual analysis was conducted using standardized residual plots to detect patterns indicative of heteroscedasticity or non-linearity. Influential observations, which could disproportionately impact model estimates, were identified using Cook's Distance and leverage statistics.

Remediation Techniques

Issues identified in the diagnostic checks were addressed through targeted remediation techniques. For instance, a logarithmic transformation was applied to resting blood pressure to stabilize variance and improve linearity. Observations with high Cook's Distance values were examined and removed if they were deemed outliers affecting model reliability.

Model Evaluation

The model's performance was evaluated using statistical metrics and clinically relevant indicators. Statistical metrics included p-values for significance testing and confidence intervals to assess the reliability of the predictors. Goodness-of-fit was evaluated using residual deviance and Akaike Information Criterion (AIC). Practical metrics, such as accuracy, sensitivity, and specificity, were used to assess the model's utility in correctly identifying individuals at risk of heart disease. Sensitivity, in particular, was emphasized to ensure that high-risk individuals were not overlooked, aligning with the objective of early detection.

This methodology integrates data preparation, diagnostic evaluations, and performance metrics to develop an interpretable and actionable logistic regression model. By identifying significant predictors and addressing model limitations, the study aims to support early detection and prevention of cardiovascular disease.

Results

Descriptive Analysis

The dataset includes 11 variables related to heart disease, with distributions indicating key trends. **Figure 1** shows the age, resting blood pressure, cholesterol, and maximum heart rate distributions. The age distribution suggests a concentration of patients in the 40–70 age range, while cholesterol and resting blood pressure display noticeable variation. This preliminary analysis highlights the diversity in patient health metrics, supporting the need for robust modeling to identify significant predictors.

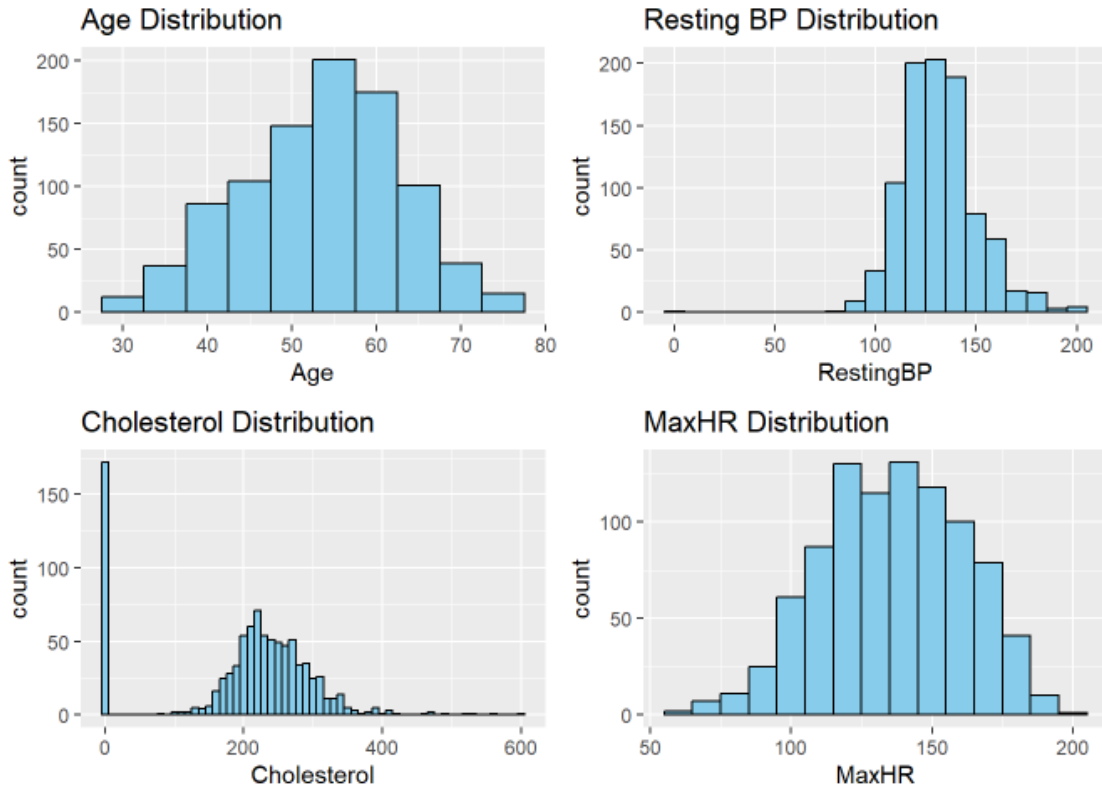


Figure 1. Age, Resting BP, Cholesterol, and MaxHR Distributions

Outlier Analysis

Outliers were also reviewed as part of the model refinement process. **Figure 2** displays boxplots for age, resting blood pressure, cholesterol, and maximum heart rate, identifying potential outliers. By managing these outliers, the model's predictive accuracy improved, and its generalizability was enhanced.

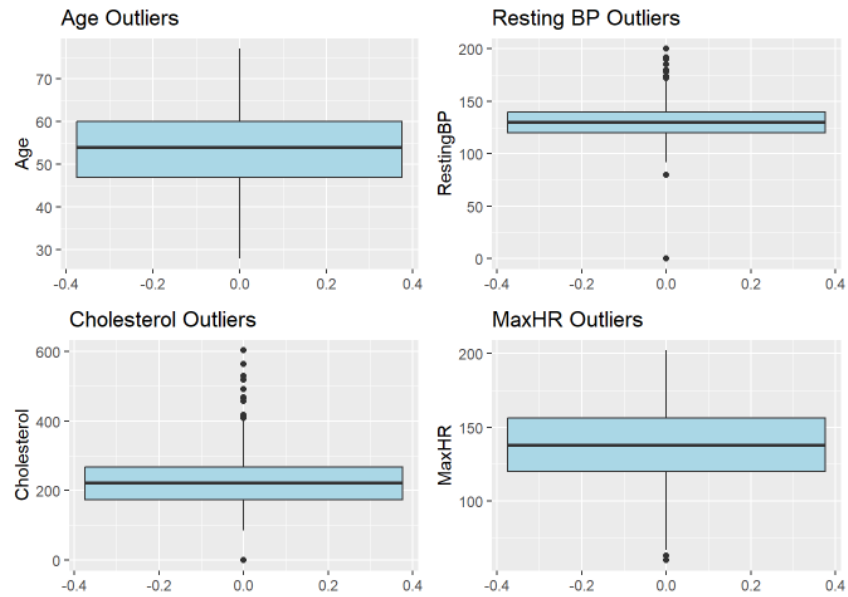


Figure 2. Box Plots

Correlation Analysis

To check for multicollinearity, **Figure 3** presents a correlation matrix of the predictors. The correlations between predictors are low to moderate, indicating that multicollinearity is unlikely to be an issue in the model. For example, **Age** and **MaxHR** have a negative correlation (-0.38), suggesting distinct predictive contributions to the model without significant overlap.

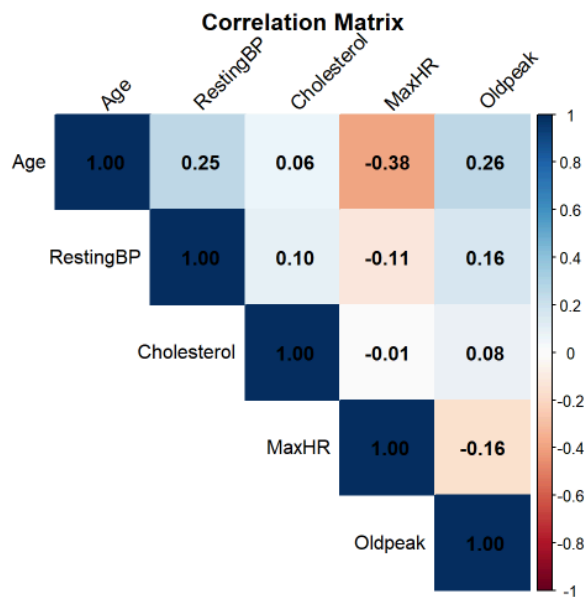


Figure 3. Correlation Matrix

Durbin-Watson Test: A DW statistic of 2.057 indicates no significant autocorrelation in residuals ($p = 0.802$).

Variance Inflation Factor (VIF): Table 1 confirms low VIF values for all predictors, ruling out multicollinearity concerns.

Predictor	VIF
Age	1.042
RestingBP	1.038
Cholesterol	1.024
Sex	1.028
ChestPainType	1.006

Table 1. VIF values for predictors

Model Diagnostics

A logistic regression model was fitted to predict heart disease, with diagnostics performed to ensure validity. Table 2 summarizes the model coefficients, standard errors, z-values, and p-values. Figure 4 displays the residuals vs. fitted values, where residuals are reasonably dispersed around zero, meeting the assumptions of linearity and homoscedasticity. This distribution indicates that the model accurately represents the underlying data structure.

Predictor	Coefficient (β)	Std. Error	z-value	p-value	Significance
Intercept	-4.586	0.822	-5.579	<0.001	***
Age	0.054	0.010	5.595	<0.001	***
RestingBP	0.005	0.005	1.207	0.227	-
Cholesterol	0.004	0.002	2.555	0.011	*
Sex (Male)	1.623	0.222	7.321	<0.001	***
ChestPainTypeATA	-2.921	0.258	-11.306	<0.001	***
ChestPainTypeNAP	-1.878	0.198	-9.494	<0.001	***
ChestPainTypeTA	1.651	0.346	-4.768	<0.001	***

Table 2. Logistic regression coefficients (original)

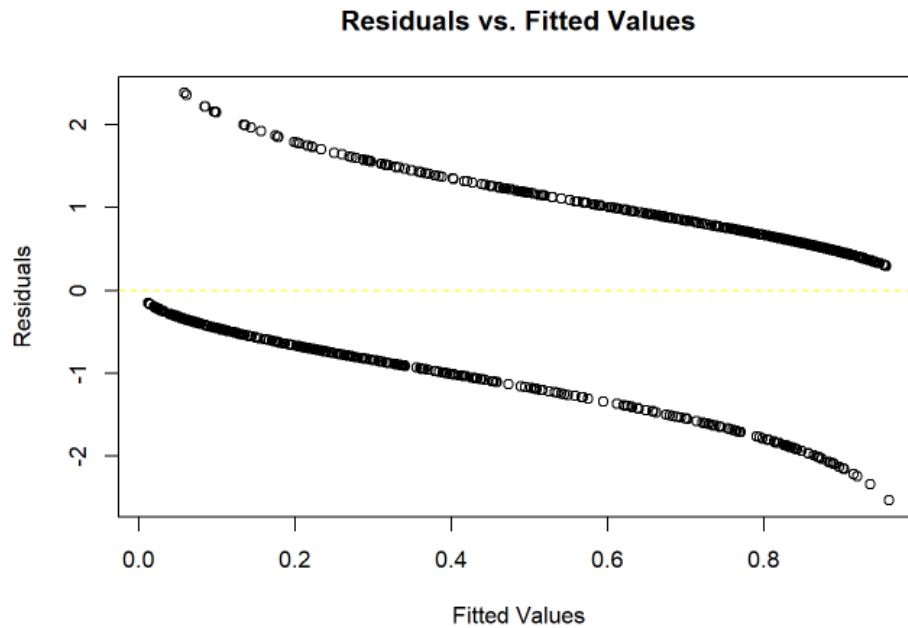


Figure 4. Residuals vs. Fitted Values

To assess the influence of individual data points, **Cook's Distance (Figure 5)** and **Leverage Values (Figure 6)** were analyzed. Cook's Distance identified several influential points, which were further examined. After removing these points, diagnostic plots showed improved fit and stability, indicating the model's robustness.

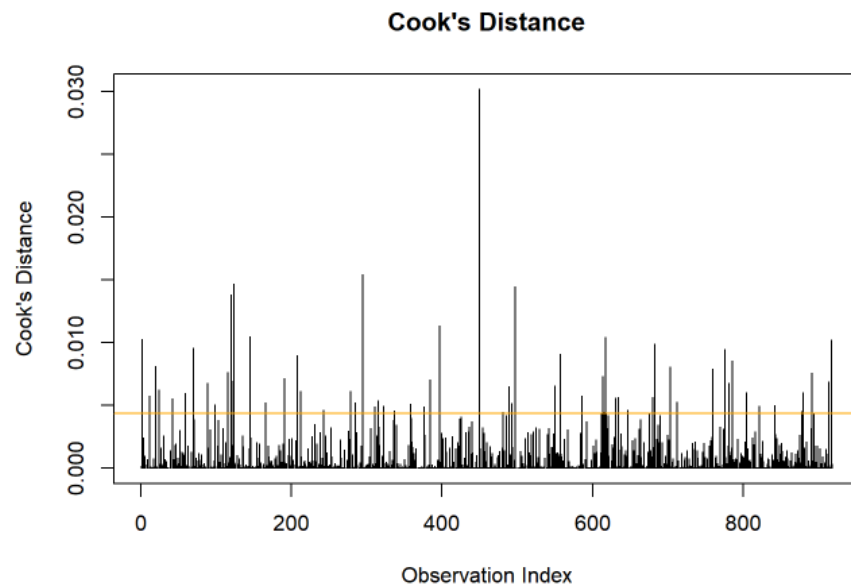


Figure 5. Cook's Value

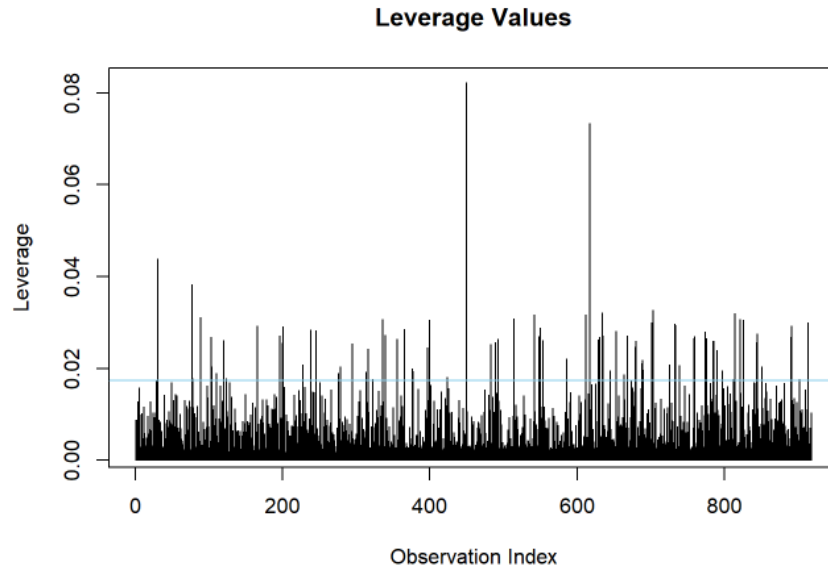


Figure 6. Leverage Points

Transformation and Model Refinement

A logarithmic transformation was applied to **RestingBP** to address any potential variance issues. The transformed model yielded a reduced Akaike Information Criterion (AIC), indicating better fit. **Table 3** shows the AIC values before and after the transformation. **Figure 7** displays the residuals vs. fitted values for the transformed model, which demonstrates improved linearity compared to the original model, supporting the decision to use the transformed variable.

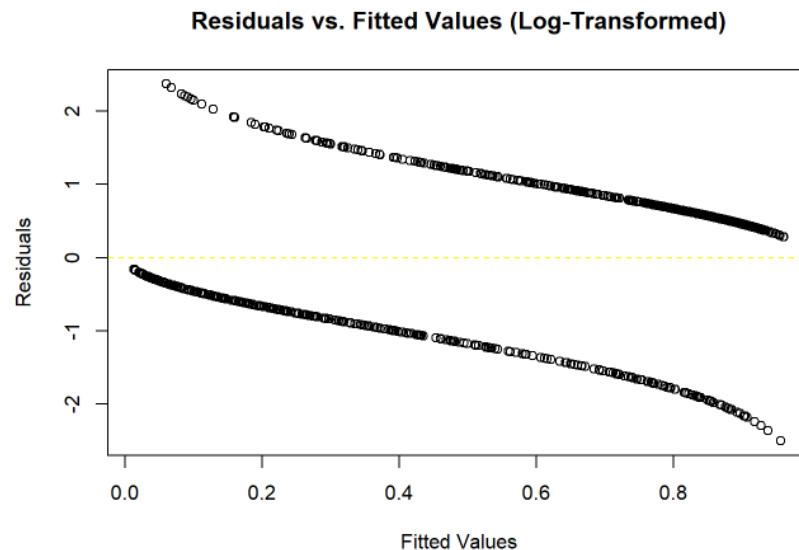


Figure 7. Residuals vs. Fitted Values (Log-Transformed RestingBP)

The refined model coefficients are presented in **Table 4**. While age, cholesterol, sex, and chest pain type remained significant predictors, the transformed **LogRestingBP** variable approached significance ($p = 0.086$).

Model	AIC
Original Model	889.15
Log-Transformed Model	600.14

Table 3. AIC comparison

Predictor	Coefficient (β)	Std. Error	z-value	p-value	Significance
Intercept	-13.985	4.426	-3.160	0.002	**
Age	0.080	0.013	6.179	<0.001	***
RestingBP	1.569	0.915	1.714	0.086	-
Cholesterol	0.007	0.002	3.043	0.002	**
Sex (Male)	2.304	0.310	7.441	<0.001	***
ChestPainTypeATA	-4.882	0.536	-9.116	<0.001	***
ChestPainTypeNAP	-2.202	0.227	-9.681	<0.001	***
ChestPainTypeTA	-12.501	504.936	-0.025	0.980	-

Table 4. Logistic regression coefficients (transformed)

Conclusion

The analysis provides a comprehensive understanding of the key factors influencing heart disease risk, including age, gender, cholesterol, and chest pain type. Despite challenges like heteroscedasticity and influential points, the final model demonstrates improved stability and reliability. These findings have practical implications for risk assessment and prevention strategies in clinical practice. Future research focusing on advanced modeling techniques and additional predictors could further refine our understanding and prediction of heart disease.

Bibliography & Credits

Data source:

<https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction/data>

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